Week 4: Mitigating Risk

Collecting Assets to Mitigate Risk?

1. If we consider more than one stock, we might be able to mitigate the risks associated with both stocks more easily.

2. If we can collect assets with different behaviors, so that if one asset goes down, the other asset will not, then we can mitigate the risk associated with the collection of these two assets.

Portfolios

Definition

A **portfolio** is simply a collection of assets, such as stocks, bonds, etc.

1. The collection of these two assets, in some proportion, is called a portfolio.

2. Portfolios allow investors to mitigate risk, to some extent.

3. Portfolio theory seeks to answer the following question: how should we invest our wealth, given that most assets are risky?

Goals

1. The goal of portfolio theory is to find the optimal collection of assets that ensures the portfolio satisfies two goals:
   - maximize the expected return
   - minimize risk
2. We measure **risk** as the standard deviation of the return on our portfolio.
3. Our goal can be stated as choosing the allocation of assets (weights), so that we simultaneously maximize expected returns, and minimize risk.

Portfolios

Investors only care about portfolio expected return and portfolio variance.

Investors like portfolios with high expected return but dislike portfolios with high return variance.

This describes the mean-variance trade-off investors face.

All else equal, investors prefer assets that have the highest expected return.

Mean-variance trade-off

However, as we have seen, assets with high expected returns tend to also have high variances.

High variance represents more uncertainty about future return and it is assumed that investors do not like uncertainty about future return. That is, we assume that investors are risk averse.

Mean-variance trade-off

Consider the following investment problem. We can invest in two assets A and B over the next year, and we want to know how much to invest in each asset to make the investor most happy given the preferences.

Let R_A and R_B denote the annual returns on assets A and B, respectively.

Portfolio

We assume that we know the means, variances, and covariances of the probability distribution of the two returns R_A and R_B:

$$\mu_A = \mathbb{E}[R_A], \quad \sigma_A^2 = \mathrm{var}(R_A),$$

$$\mu_B = \mathbb{E}[R_B], \quad \sigma_B^2 = \mathrm{var}(R_B),$$

$$\sigma_{AB} = \mathrm{cov}(R_A, R_B), \qquad \rho_{AB} = \frac{\sigma_{AB}}{\sigma_A \sigma_B}.$$

Expected returns and variances

The expected returns, μ_A and μ_B, are our best guesses for the annual returns on each of the assets.

The variances, σ_A^2 and σ_B^2, provide measures of the uncertainty associated with these annual returns. We can also think of the variances as measuring the risk associated with the investments.

Assets with high return variability (or volatility) are often thought to be risky, and assets with low return volatility are often thought to be safe.

Covariance and correlation coefficient

The covariance σ_{AB} gives us information about the direction of any linear dependence between returns.

- If $\sigma_{AB} > 0$ then the two returns tend to move in the same direction;
- if $\sigma_{AB} < 0$ the returns tend to move in opposite directions;
- if $\sigma_{AB} = 0$ then the returns tend to move independently.

The correlation coefficient, ρ_{AB}, measures the strength of the dependence between the returns.

If ρ_{AB} is close to one in absolute value then returns mimic each other extremely closely, whereas if ρ_{AB} is close to zero then the returns may show very little relationship.

This table gives annual return distribution parameters for the two assets A and B.

μ_A	μ_B	σ_A^2	σ_B^2	σ_A	σ_B	σ_{AB}	ρ_{AB}
0.175	0.055	0.067	0.013	0.258	0.115	-0.005	-0.164

Asset A is the high risk asset with an annual return of $\mu_A = 17.5\%$ and annual standard deviation of $\sigma_A = 25.8\%$.

Asset B is a lower risk asset with annual return $\mu_B = 5.5\%$ and annual standard deviation of $\sigma_B = 11.5\%$.

These assets are assumed to be slightly negatively correlated with correlation coefficient $\rho_{AB} = -0.164$.

Covariance

Given the standard deviations and the correlation, the covariance can be determined from

$$\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B = (-0.164)(0.258)(0.115) = -0.00487.$$

The investment in the two assets forms a *portfolio*, and the shares x_A and x_B are referred to as *portfolio shares* or *portfolio weights*.
The return on the portfolio is given by:

$$R_p = x_A R_A + x_B R_B.$$

with $x_A \geq 0$ and $x_B \geq 0$. Since all wealth is put into the two investments it follows that $x_A + x_B = 1$.

$$\mu_p = \mathbb{E}[R_p] = x_A \mu_A + x_B \mu_B,$$

$$\sigma_p^2 = \mathrm{var}(R_p) = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2 x_A x_B \sigma_{AB},$$

$$\sigma_p = \sqrt{x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2 x_A x_B \sigma_{AB}}.$$

The results in the previous slide are so important to portfolio theory.

For μ_p, we have:

$$E[R_p] = E[x_A R_A + x_B R_B] = x_A E[R_A] + x_B E[R_B] = x_A \mu_A + x_B \mu_B,$$

For $\mathrm{var}(R_p)$, we have:

$$\begin{aligned}
\mathrm{var}(R_p) &= E[(R_p - \mu_p)^2] = E[(x_A(R_A - \mu_A) + x_B(R_B - \mu_B))^2] \\
&= E[x_A^2(R_A - \mu_A)^2 + x_B^2(R_B - \mu_B)^2 + 2x_A x_B(R_A - \mu_A)(R_B - \mu_B)] \\
&= x_A^2 E[(R_A - \mu_A)^2] + x_B^2 E[(R_B - \mu_B)^2] \\
&\quad + 2x_A x_B E[(R_A - \mu_A)(R_B - \mu_B)] \\
&= x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \sigma_{AB}.
\end{aligned}$$

Notice that the variance of the portfolio is a weighted average of the variances of the individual assets plus two times the product of the portfolio weights times the covariance between the assets.

If the portfolio weights are both positive then a positive covariance will tend to increase the portfolio variance, because both returns tend to move in the same direction, and a negative covariance will tend to reduce the portfolio variance as returns tend to move in opposite directions.

Thus finding assets with negatively correlated returns can be very beneficial when forming portfolios because risk, as measured by portfolio standard deviation, can be reduced.

Or, if we cannot do that, at least avoid shares whose returns are very highly positively correlated with each other.

Two asset portfolios

Consider creating an equally weighted portfolio with $x_A = x_B = 0.5$ using the asset information in the Table.

The expected return, variance and standard deviation of this equally weighted portfolio are:

$$\mu_p = (0.5) \cdot (0.175) + (0.5) \cdot (0.055) = 0.115$$

$$\begin{aligned} \sigma_p^2 &= (0.5)^2 \cdot (0.067) + (0.5)^2 \cdot (0.013) + 2 \cdot (0.5)(0.5)(-0.00487) \\ &= 0.01751 \end{aligned}$$

$$\sigma_p = \sqrt{0.01751} = 0.1323$$

Two asset portfolios

This portfolio has expected return half-way between the expected returns on assets A and B, but the portfolio standard deviation is less than σ_A.

This reflects risk reduction via diversification. Here, diversification means investing wealth in two assets instead of investing wealth in a single asset.

Normally Distributed

If the distribution of assets returns is jointly normal then the probability distribution of returns is completely characterized by means, variances, and covariances.

In the above example, we assume that the returns R_A and R_B are jointly normally distributed.

One Risky Asset, One Riskless Asset

One Risky Asset, One Riskless Asset

First, consider optimal portfolio allocation in the context of one risky asset (R), and one risk-free asset R_f:

The probability of default on a 1-month U.S. treasury bill is so small that these assets are effectively riskless.

Main Problem:

Construct a ***portfolio*** *for R and* R_f *that we will hold for a single time period (whatever this period may be). Let* μ_f *denote the expected return on the risk-free asset and* μ_R *the expected return on the risky asset.* ***Then, our goal is to find an allocation rule*** $w \in [0, 1]$.

Portfolio Allocation

Returns on such a portfolio can be represented as

$$R_p = wR + (1 - w)R_f.$$

Re-arrange gives

$$\begin{aligned} R_p &= wR + (1 - w)\,R_f \\ &= R_f + w\left(R - R_f\right) \end{aligned}$$

The quantity $R - R_f$ is called the excess return (over the return on T-bills) on the risky asset.

Expected Return

$$R_p = wR + (1 - w)R_f.$$

The expected return on the portfolio is

$$\mathbb{E}[R_p] = w\mu_R + (1 - w)\mu_f$$

or

$$\mathbb{E}[R_p] = \mu_f + w(\mu_R - \mu_f)$$

where the quantity $\mu_R - \mu_f$ is called the expected excess return or **risk premium** on the risky asset.

The risk premium is typically positive indicating that investors expect a higher return on the risky asset than the safe asset (otherwise, why would investors hold the risky asset?).

Variance

$$R_p = wR + (1 - w)R_f.$$

The variance of the portfolio is

$$\mathbb{V}[R_p] = w^2\sigma_R^2 + (1 - w)^2\sigma_{R_f}^2$$

However, if the riskless asset indeed has no risk, then it is constant and $\mathbb{E}[R_f] = \mu_f$, while $\mathbb{V}[R_f] = 0$, and

$$\mathbb{V}[R_p] = w^2\sigma_R^2.$$

Optimal Allocation

- To decide what proportion w of one's wealth to invest in the risky asset R we must either choose:

  1. the expected return $\mathbb{E}[R_p]$ one wishes to obtain, or

  2. the amount of risk $\mathbb{V}[R_p]$ with which one is willing to live with.

- Once either $\mathbb{E}[R_p]$ or $\mathbb{V}[R_p]$ is chosen, we can determine w.

Optimal Allocation

Example

Suppose we wish to obtain an expected return on the portfolio of $\mathbb{E}[R_p] = .1$, what should w be?

In this case, we just need to solve the equation

$$\mathbb{E}[R_p] = .1 = w\mu_R + (1-w)\mu_f,$$

which implies that

$$w = \frac{.1 - \mu_f}{\mu_R - \mu_f}.$$

Optimal Allocation

1. More generally, for an expected return on the portfolio, we must invest a percentage of our wealth equal to

$$w = \frac{\mu_{R_p} - \mu_f}{\mu_R - \mu_f}$$

in the risky asset.

2. This allocation for w says nothing about the risk inherent in this choice.

Example

Suppose we are not concerned with maximizing the expected return, but are concerned with minimizing risk. Say we are only willing to accept a risk of $\sigma_{R_p} = 5\%$. Then, what is the optimal choice of w?

- This can be obtained by solving

$$\mathbb{V}[R_p]^{1/2} = .05 = w\sigma_R,$$

- $\Longrightarrow$

$$w = .05/\sigma_R.$$

- More generally, for a level of risk given by σ_{R_p}, we can choose w according to this level of risk by choosing

$$w = \sigma_{R_p}/\sigma_R.$$

Optimal Allocation

- An allocation using σ_{R_p} says nothing about the expected return associated with this portfolio.
- This can be found by replacing w with $w = \sigma_{R_p}/\sigma_R$.
- $\implies$ expected returns

$$\mu_{R_p} = \mu_f + \frac{\mu_R - \mu_f}{\sigma_R} \cdot \sigma_{R_p} \tag{1}$$

Optimal Allocation

Equation (1) describes a straight line in (μ_p, σ_p)-space with intercept μ_f and slope

$$\frac{\mu_R - \mu_f}{\sigma_R}.$$

This line characterizes the linear risk-return tradeoff of portfolios of T-Bills and a risky asset.

This line has a positive slope provided the risk premium is positive.

Optimal Allocation

This line described by (1) is often called the Capital Allocation Line (CAL) as it describes how investment capital is allocated between T-Bills and the risky asset.

Points on the line close to the y-axis represent portfolios mostly allocated to T-Bills.

The slope of the CAL is called the Sharpe ratio (SR) (named after the economist William Sharpe). From equation (1) we have

$$SR = \frac{\mu_R - \mu_f}{\sigma_R}.$$

Example

Consider creating portfolios of asset A and T-bills and asset B and T-Bills with $\mu_f = 0.03$.

Using the data in the Table, we calculate the Sharpe ratio for asset *A* and *B*.

The portfolios which are combinations of asset *A* and T-bills have expected returns uniformly higher than the portfolios consisting of asset *B* and T-bills.

Example

This occurs because the Sharpe ratio for asset A is higher than the ratio for asset B:

$$SR_A = \frac{\mu_R^A - \mu_f}{\sigma_R^A} = \frac{0.175 - 0.03}{0.258} = 0.562,$$

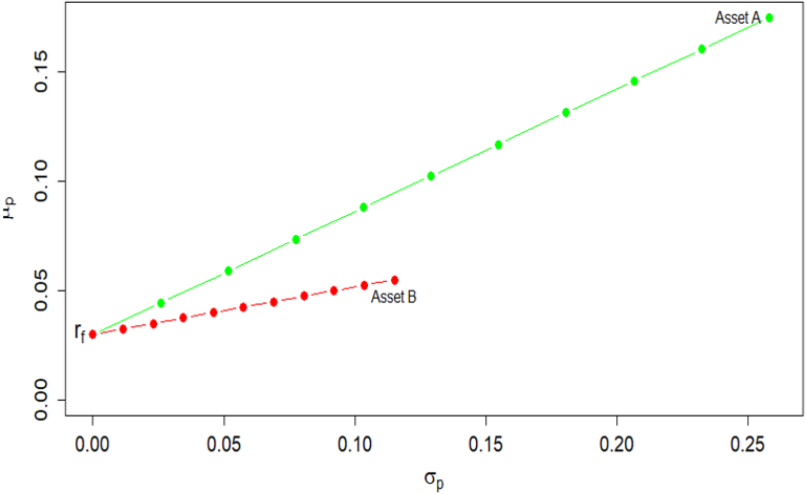
$$SR_B = \frac{\mu_R^B - \mu_f}{\sigma_R^B} = \frac{0.055 - 0.03}{0.115} = 0.217.$$

Hence, portfolios of asset A and T-bills are *efficient* relative to portfolios of asset B and T-bills.

Example

Efficient portfolios are those with the highest expected return for a given level of risk.

Inefficient portfolios are then portfolios such that there is another feasible portfolio that has the same risk (σ_{R_p}) but a higher expected return (μ_{R_p}).


Asset A
Asset B
r_f
0.00
0.05
0.10
0.15
0.20
0.25
σ_p
μ_p

Example

In the Figure, the point labelled "r_f" represents a portfolio that is 100% invested in T-bills.

The points labelled "Asset A" and "Asset B" represent portfolios that are 100% invested in assets A and B, respectively.

A point half-way between "r_f" and "Asset A" is a portfolio that is 50% invested in T-bills and 50% invested in asset A.

Example

The previous example shows that the Sharpe ratio can be used to rank the risk return properties of individual assets.

Assets with a high Sharpe ratio have a better risk-return tradeoff than assets with a low Sharpe ratio.

Accordingly, investment analysts routinely rank assets based on their Sharpe ratios.

Portfolio Allocation: Two Risky Assets

Portfolio Allocation: Two Risky Assets

- Now, choose a portfolio formed from two risky assets R_1 and R_2.

- All portfolios are given by

$$R_p = wR_1 + (1 - w)R_2.$$

- We consider that, for $j = 1, 2$, $\mathbb{E}[R_j] = \mu_j$ and $\mathbb{V}[R_j] = \sigma_j^2$ with

$$\sigma_{12} = \mathrm{Cov}(R_1, R_2) = \mathbb{E}[(R_1 - \mathbb{E}[R_1])(R_2 - \mathbb{E}[R_2])].$$

Two Risky Assets

- Expected return on the portfolio is

$$\mu_{R_p} = \mathbb{E}[R_p] = w\mu_{R_1} + (1-w)\mu_{R_2}.$$

- The (squared) portfolio risk is given by

$$\begin{aligned}\sigma_{R_p}^2 &= w^2\mathbb{V}[R_1] + (1-w)^2\mathbb{V}[R_2] + 2w(1-w)\mathrm{Cov}(R_1, R_2)\\ &= w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}\end{aligned}$$

Main Problem:

Find an allocation rule, w that results in a portfolio that minimizes risk. This is equivalent to solving the following optimization problem

$$\min_{w\in[0,1]} \sigma_{R_p}^2.$$

Optimization problem

- This is an optimization problem in w:

$$\min_{w \in [0,1]} \sigma_{R_p}^2$$

with solution

$$w = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}.$$

Derivation of the optimization problem

We derive the optimization problem as follows:

The allocation rule can be obtained by differentiating, with respect to w, $\sigma_{R_p}^2$:

$$\begin{aligned}\frac{d\sigma_{R_p}^2}{dw} &= 2w\sigma_1^2 - 2(1-w)\sigma_2^2 + 2(1-w)\sigma_{12} - 2w\sigma_{12} \\ &= 2w(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - 2\sigma_2^2 + 2\sigma_{12} = 0\end{aligned}$$

We can directly solve these first-order conditions to obtain

$$\hat{w} = \frac{\sigma_2^2 - \sigma_{12}}{(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})},\ 1 - \hat{w} = \frac{\sigma_1^2 - \sigma_{12}}{(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})}.$$

- To make sure this is indeed a minimum, we must look at second order conditions:

$$\frac{d\sigma_{R_p}^2}{dw} = 2w(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - 2\sigma_2^2 + 2\sigma_{12}$$

$$\Longrightarrow \frac{d^2\sigma_{R_p}^2}{dw^2} = 2(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) > 0.$$

- Yes, indeed it is a minimum.
- The same result is obtained when using the method of Lagrange multipliers.

Example

Using the data in the earlier Table, we have:

$$x_A^{\min} = \frac{\sigma_B^2 - \sigma_{AB}}{(\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB})}$$

or

$$x_A^{\min} = \frac{0.01323 - (-0.004866)}{0.06656 + 0.01323 - 2(-0.004866)} = 0.2021,$$

and

$$x_B^{\min} = 0.7979$$

This optimal portfolio is called the **Global Minimum Variance portfolio** because it has the smallest variance/volatility among all feasible portfolios.

Global Minimum Variance (GMV) portfolio

The expected return, variance and standard deviation of this GMV portfolio are:

$$\mu_p = (0.2021) \cdot (0.175) + (0.7979) \cdot (0.055) = 0.07925$$

$$\begin{aligned} \sigma_p^2 &= (0.2021)^2 \cdot (0.067) + (0.7979)^2 \cdot (0.013) \\ &\quad + 2 \cdot (0.2021)(0.7979)(-0.00487) \\ &= 0.00975 \end{aligned}$$

$$\sigma_p = \sqrt{0.00975} = 0.09782.$$

Global Minimum Variance portfolio

The GMV portfolio is obtained as

$$\begin{aligned} R_p &= \hat{w} R_1 + (1 - \hat{w}) R_2 \\ &= R_2 + (R_1 - R_2)\hat{w} \\ &= R_2 + (R_1 - R_2)\frac{\sigma_2^2 - \sigma_{12}}{\sigma_2^2 + \sigma_1^2 - 2\sigma_{12}} \end{aligned}$$

Global Minimum Variance Portfolio

The expected return of the GMV portfolio is given by

$$
\begin{aligned}
\mu_p &= \hat{w}\mu_1 + (1-\hat{w})\mu_2 \\
&= \mu_2 + (\mu_1 - \mu_2)\hat{w} \\
&= \mu_2 + (\mu_1 - \mu_2)\frac{\sigma_2^2 - \sigma_{12}}{\sigma_2^2 + \sigma_1^2 - 2\sigma_{12}}
\end{aligned}
$$

Conditional Optimal Portfolio Allocation

- While the above may not look completely familiar, you have seen this formula before.

- Define

$$X = R_2 - R_1 \text{ and } Y = R_2$$

- Note

$$\mathrm{Cov}(Y, X) = \sigma_2^2 - \sigma_{12}$$

and

$$\mathbb{V}[X] = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}.$$

Conditional Optimal Portfolio Allocation

- We can rewrite μ_p as

$$\mu_p = \mathbb{E}[R_p] = \mu_y - \mu_x \beta_1, \qquad \beta_1 = \frac{\mathrm{Cov}(Y, X)}{\mathbb{V}(X)}.$$

- This seems very similar to the **population** regression line from the simple linear model.

- If this problem could be interpreted as a linear regression model, it would make estimation of w very simple.

Conditional Optimal Portfolio Allocation

- Recall: **population** linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where $\mathbb{E}[\epsilon|X] = 0$ and

$$\beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X]$$

$$\beta_1 = \frac{\mathbb{E}[(Y - \mathbb{E}[Y])(X - \mathbb{E}[X])]}{\mathbb{V}[X]}.$$

- From the above definitions: for $X = R_2 - R_1$ and $Y = R_2$:

$$\beta_0 = \mu_2 - \beta_1(\mu_2 - \mu_1),$$

and

$$\beta_1 = \frac{\mathrm{Cov}(X, Y)}{\mathbb{V}[X]} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_2^2 + \sigma_1^2 - 2\sigma_{12}} = w.$$

Therefore, minimum variance portfolio optimization can be re-stated as the **population** linear regression model

$$
\begin{aligned}
R_2 &= w\mu_1 + (1-w)\mu_2 + w(R_2 - R_1) + \epsilon \\
R_2 &= \mu_p + w(R_2 - R_1) + \epsilon \\
&= \beta_0 + \beta_1(R_2 - R_1) + \epsilon \\
&\Longrightarrow \\
Y &= \beta_0 + \beta_1 X + \epsilon.
\end{aligned}
$$

Portfolio weights that minimize the risk can be obtained by OLS:

1. "Outcome": variable R_2;
2. "Covariate": $R_2 - R_1$.

Given a time series of returns on two assets $\{R_{1t}, R_{2t}\}_{t=1}^{T}$.

We can estimate the optimal portfolio weights by solving

$$\min_{\beta_0,\beta_1} \sum_{t=1}^{T} (R_{2t} - \beta_0 - \beta_1 (R_{2t} - R_{1t}))^2.$$

Define $y_t = R_{2t}$ and $x_t = R_{2t} - R_{1t}$. OLS solutions:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$
$$\hat{\beta}_1 = \frac{\sum_{t=1}^{T} (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^{T} (x_t - \bar{x})^2}$$

$\hat{\beta}_1$ is the optimal portfolio weight w. The one associated with minimizing the variance of the portfolio.

Example

Let us figure out the GMV portfolio that is composed of IBM and BHP.

Let $r_{1,t}$ be the **log return** on IBM and $r_{2,t}$ be the **log return** on BHP.

Fitted regression line:

$$\hat{r}_{2,t} = \hat{\beta}_0 + \hat{\beta}_1(r_{2,t} - r_{1,t})$$
$$\hat{r}_{2,t} = 0.0001 + 0.80(r_{2,t} - r_{1,t})$$

Should invest nearly 80% of wealth in IBM, and only 20% in BHP.

Capital Asset Pricing Model (CAPM)

CAPM

One of the most important models in finance is the Capital Asset Pricing Model (CAPM).

So far, the focus has been on an individual investor.

How much they should include a stock in their portfolio is determined by the stock's contribution to the risk (standard deviation) and return (mean) of the portfolio.

As such, what matters are the correlations between this stock and all other existing stocks in the portfolio.

In the CAPM, the focus is on the entire economy. What matters is a stock's **correlation** with the market portfolio R_M.

Key ingredients of the CAPM

- The market risk premium: defined as the expected return of the market portfolio R_M in excess of the riskfree rate r_f:

$$\mathbb{E}(R_M) - r_f$$

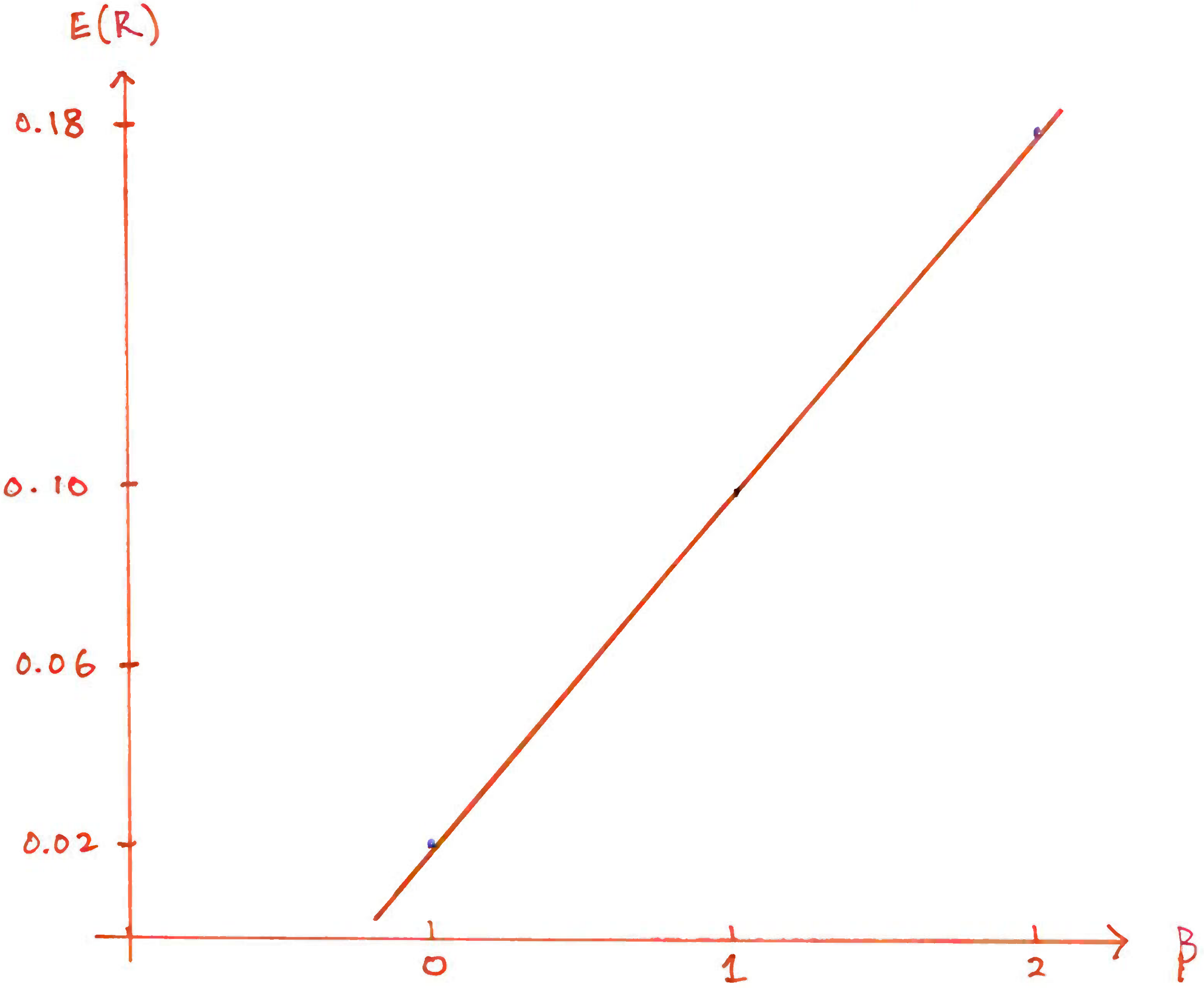
- Beta: The risk of an individual stock, say A, is measured by its exposure to the market risk:

$$\beta^A = \frac{\text{cov}\left(R_A, R_M\right)}{\text{var}\left(R_M\right)}$$

- **The pricing equation**: The reward is proportional to risk:

$$\mathbb{E}(R_A) - r_f = \beta^A \times \left(\mathbb{E}(R_M) - r_f\right)$$

$$E(R) = r_f + \beta\left(E(R_m) - r_f\right)$$



Idiosyncratic risk

Idiosyncratic risk derives from the fact that many of the dangers that surround an individual company are peculiar to that company and, perhaps, its immediate competitors.

Idiosyncratic risk can therefore be eliminated by holding a well-diversified portfolio.

But there is also some risk that cannot be avoided no matter how much one diversifies. This risk is generally known as market risk.

Market risk

Market risk is sometimes called the **systematic risk** .

Market risk derives from the fact that there are other economywide and global dangers and opportunities that are faced by all businesses.

The fact that stocks have a tendency to ”move together” reflects the presence of market risk, risk that cannot be eliminated through diversification.

In CAPM, market risk is captured by **beta** (β).

1990 Nobel Prize, Harry Markowitz (Week 4)